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Common Grocery Self-Service Checkout Failures That Customers Can Resolve Without Employee Assistance

Insights from Store Employees

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Abstract

This thesis explores the common challenges customers face when using self-service checkouts (SCOs) in Finnish grocery retail, a key application of the broader concept of self-service technologies (SSTs). It examines how these issues might be addressed to reduce employee intervention. Through qualitative interviews with store employees, the study identifies recurring service breakdowns and highlights how both user behavior and system design contribute to checkout friction.

The findings suggest that many SCO-related problems come from interface limitations, unclear instructions, and the complexity of retail processes. Employees offered several practical suggestions for improving the self-checkout experience, including technical enhancements, interface adjustments, and clearer communication strategies. These improvements aim to make SCOs more intuitive and autonomous for users while easing the burden on employees.

By focusing on the Finnish context, this study contributes to broader discussions on SSTs adoption, emphasizing the importance of employee insight and localized design in enhancing the customer experience.

Keywords Self-Service Checkout (SCO), Self-Service Technology (SST), Service Failures, Employee Intervention

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Preface and acknowledgements

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Espoo, 02 June 2025

Linh Duong

1 Introduction

1.1 Background

Improving productivity in the service sector has been important for organizations to enhance their customer experience and operational efficiency (Parasuraman, 2010). Technology exists everywhere in the service sector and is the primary source to improve productivity (Huang & Rust, 2017). In service sector, the main role of technology is to provide communication and information (Huang & Rust, 2017). Client interface is one of the main venues for service innovations (Miles, 2008) when many companies have been providing greater levels of self-service for clients (B. S. Galdolage, 2022). The service encounters that have been traditionally handled by service employees in physical service premises are now reduced with self-service technology (SST) (Tapani Rinta-Kahila, 2013). Customers can interact with technological interfaces to handle service encounters on their own (B. S. Galdolage, 2022). SST has been altering the nature of service encounters radically.

There are many SST applications such as bank ATMs, self-checkin at airports and hotels, self-service checkouts in grocery stores, and web-based services (.B. Lee & Cranage, 2018). From customers' perspective, SSTs empower them with service speed and better customization; on the other hand, SSTs improve efficiency, reliability and reduce operating cost for service providers (Meuter et al., 2005). This study focuses on one of the SST applications in retailing industry, self-service checkouts (SCOs). With SCOs, customer experience has improved since it is easier and faster for customers to order and pay for retail products (Wisastra et al., 2024). By utilizing SCOs in stores, retail enterprises have saved personnel costs, better managed demand fluctuations with real-time transaction data, and increased efficiency in the checkout process (Tapani Rinta-Kahila, 2013).

Despite its advantages, SST can have many failures because the increased customer responsibility in its usage expands the risk of errors (Collier et al., 2017). There are two major types of SST failures: technology and customer-induced failures (Dabholkar & Spaid, 2012; Dao & Theotokis, 2021; Meuter et al., 2000). Due to these failures, customers can perceive the service provider negatively (Collier et al., 2017). Key reasons for satisfaction and dissatisfaction with SSTs are identified, including technology failure, process failure, poor design, and customer-induced failure (Meuter et al., 2000).

SCOs, as a prominent and widely adopted form of SST in the retail sector, are similarly affected by these issues. Technical malfunctions and customer-related mistakes can disrupt the checkout experience. SCO failures not only cause frustration for customers but may also reduce their trust in the retailer and discourage future use of the technology. Therefore, it is crucial for retailers to proactively identify potential failures in SCOs and implement effective recovery strategies to minimize disruptions, enhance customer satisfaction, and maintain trust in the self-service experience.

1.2 Research Problem and Objectives

SCOs have become an important part of modern retail operations, which provide customers with an autonomous shopping experience (Le et al., 2022). However, despite their many advantages, SCOs are still prone to failures that often require assistance from store employees. These failures can disrupt the shopping experience, decrease store efficiency, and increase employee workload. While some failures are technological in nature, others can be due to customer mistakes or confusion. Even with non-technical issues, customers often need assistance from employees, which can be frustrating, as avoiding human interaction is one of the main reasons customers choose to use SCOs (Meuter et al., 2000).

The problem is that many of these failures could potentially be resolved by customers themselves with the technology, user interfaces, or instructions

that are designed to better support customer self-resolution (Mattila & Cho, 2011). Therefore, to improve the overall efficiency of SCOs, it is important to identify the common types of failures that customers can resolve independently.

The objective of this research is to identify common SCO failures that could be resolved by customers themselves, therefore, reducing the need for employee intervention and improving overall store efficiency. Moreover, the study focuses on technological improvements and solutions that enable customers to resolve issues independently. The insights for this study are gained from the employees' perspective at S Group and K Group, two major grocery retailers in Finland. Overall, this research aims to answer the following questions:

- What common problems do customers face with SCOs that need employee help?
- How can the design, technology, or instructions of SCOs be improved to help customers resolve these failures independently, reducing the need for employee intervention?

Ultimately, the objective is to improve the SCO experience with customer autonomy, which improves operational efficiency and enhances the customer experience.

1.3 SCOs in Finnish Grocery Retail

According to Finnish Grocery Trade Association, in 2024, Finland's grocery retail market is controlled by two major players: S Group, with a 48.8% market share, and K Group, holding 33.7%. In 2024, grocery retail sales in Finland were 23.5 billion euros in total. That year, S Group was the only one to increase its market share. S Group and K Group together accounted for 82.5% of Finland's grocery retail market, while the remaining share was

distributed among smaller players such as Lidl Finland (9.4%) and Tokmanni Group (3.3%) (Finnish Grocery Trade Association, n.d.)

SCOs were first tested in Finland in the 1990s by S Group in some S-Market stores; this is the earliest adoption of the technology. However, it was later discontinued because both the technology and customers were not yet prepared to adopt self-service for the checkout process (Tapani Rinta-Kahila, 2013). SCOs reappeared in Finland in 2012 through a pilot program, led by S Group and K Group. The slow adoption of SCOs may be due to Finland's highly concentrated retail market, where the dominance of these two players reduces the motivation to invest in new technologies (ibid). By the 2020s, SCOs had become a standard feature in Finnish grocery stores. They were widely adopted by major retailers like S Group, K Group, Minimani, Lidl (Elina Kaakinen, 2021) and the leading kiosk chain R-Kioski (Jari Pietiläinen, 2023).

In 2012, when SCOs were reintroduced in Finnish grocery stores, the Finnish Commerce Federation expressed skepticism with concerns about the high cost of the technology and the potential for increased theft (Helsingin Sanomat, 2012). While SCOs have become a common feature in Finnish stores, hesitation persists when it comes to fully automated shopping experiences. Representatives from Service Union United (PAM) (Elina Kaakinen, 2021) and K Group (Jari Pietiläinen, 2023) have expressed doubts about whether customers are ready for a completely self-service checkout experience. While customers often express a desire for employees to assist them with any issues (Dabholkar & Spaid, 2012), there may be a need to empower them to solve some problems independently, which could pave the way for more widespread use of SCOs in the future. This highlights the need to help customers resolve common SCO issues independently by reducing reliance on employees and supporting the move toward fully self-service checkouts.

1.4 Structure of the thesis

This study is structured as follows: Chapter 1 provides an overview of the research, including background, motivation, scope, research gaps, and objectives, along with a discussion of SCOs in Finland's retail environment. Chapter 2 reviews relevant literature and key concepts, covering service quality management, the evolution of SSTs, self-service recovery, and customer perspectives on SST failures. Chapter 3 outlines the research methodology, including the design, setting, data collection, and analysis approach. Chapter 4 presents the findings and analysis, identifying common SCO failures based on employee feedback and proposing technological and process-based solutions. Chapter 5 discusses the main findings, their theoretical and managerial implications, and outlines limitations and future research directions. The appendix includes the interview questions used in the study.

2 Literature review

This literature review aims to learn more about existing research on key aspects of SCOs in grocery retail. It explores essential concepts and highlights the significance of this thesis's empirical study. By examining literature on service quality management, SSTs, self-service recovery, and customer perspectives, this chapter provides a background for further research and develops a conceptual framework.

The literature review is divided into four main areas. First, it examines service quality management to define key concepts relevant to retail settings and explore frameworks used to measure and improve service performance in grocery environments. Second, it reviews research on SSTs, focusing on their adoption, usability, and influence on customer behavior, as well as their role in improving operational efficiency. Third, the section explores self-service recovery literature to understand how errors and failures in SSTs are handled, including strategies for error detection and customer support. Fourth, it analyzes customer attitudes and reactions to SST issues, considering factors such as ease of use, satisfaction, perceived service quality, and how customer expectations shape adoption and continued usage.

2.1 Service Quality Management

2.1.1 Definitions of service quality

Service quality is an evaluation from customers when they compare their expectations of a service with their actual service experience (Grönroos, 1984). Service quality is determined by two factors: expected service and perceived service. A consumer's prior experience with a service can shape their expectations, while perceived service reflects their actual experience and their evaluation of the service (ibid). Service quality is also defined as the gap between customers' expectations and their perceptions of service performance (Parasuraman et al., 1988). Service quality is measured based on the extent and direction of this discrepancy (ibid). (Grönroos, 1984) also

proposed that service quality can be divided into two aspects: technical quality, referring to what is delivered, and functional quality, which focuses on how the service is provided. Kang (2006) further emphasized that these two dimensions are hierarchically integrated, with functional quality often playing a more dominant role in customer perceptions, especially when technical differences are not easily evaluated by the customer.

Employees play an important role in ensuring the quality of services (Dotchin & Oakland, 1994). This is mostly due to the interaction between customers and service employees during service delivery. The need for customer-employee interaction remains important despite the technology used in service delivery (Quinn et al., 1988) and efforts to reduce the interaction (Schmenner, 1986). When face-to-face interactions become less frequent, the quality of these encounters becomes even more important. These moments of direct contact often occur when systems fail, which means that how they are handled the failure can have a significant impact on customer satisfaction (Johnston, 1989). A well-handled interaction can help restore trust, while poor service during these critical moments can leave a lasting negative impression (Dotchin & Oakland, 1994).

Customer action is essential for the success of service operations, particularly when responsibilities shift from service providers to customers. To ensure a smooth transition, it is crucial to set appropriate expectations for customers before implementing such changes. In addition, processes should be straightforward and relevant to customers. To engage customer more in the process, employees must be skilled in demonstrating new procedures and technologies (Johnston, 1989). The need to guide customer action and the supporting role of employees in this process will be examined further in the empirical part of this study.

These definitions of service quality provide a foundation for evaluating customer experiences with self-service kiosks. In self-service environments,

where direct employee interaction is reduced, perceived service quality is measured with system usability and failure recovery strategies (H.-J. Lee, 2015). How stores handle failure at SCOs can either restore or damage customer trust; this emphasizes the need for effective service recovery strategies (B. S. Galdolage, 2022).

Service quality influences both satisfaction and long-term customer behavior. The next section explores customer loyalty, its link to service quality, and its implications for SSTs.

2.1.2 Customer loyalty

Customer perception of service quality is widely acknowledged as a key driver of customer satisfaction, loyalty, and competitive advantage. Research suggests a causal relationship among these factors with service quality as the initiating element and customer satisfaction as a mediator in this process (Fornell et al., 1996). Service quality has been identified as a key factor influencing customer behavior, including the likelihood of recommending a service, making repeat purchases, switching providers, or having complaints (Bitner, 1990). The strong correlation between service quality and repurchase behavior was also supported by other studies Woodside et al., 1989 and Dabholkar et al., 1996.

Past literature has presented several definitions of customer loyalty. Loyalty is often described as an attitude and behavior commitment such as repeat purchases or a positive emotional connection to a brand (Funke & Day, 1976). It involves continued engagement with a product or service over time despite price fluctuations (Lewis & Soureli, 2006) (Wangenheim & Bayon, 2004), and a higher likelihood of recommending the brand to others through word-of-mouth (Bowen & Shoemaker, 2003). Customer loyalty is also viewed as a relational construct, where repeat behavior and psychological attachment reinforce each other (Jarvis & Wilcox, 1977). In summary, customer loyalty is generally defined as the repeated purchase of goods or services over time

with a positive attitude toward and continued support for the company providing them.

In the context of SCOs, customer loyalty is very relevant due to the lack of direct employee interaction. This nature of the interaction shifts the responsibility of service quality perception to the technology. A smooth, efficient, and error-free SCO experience can enhance customer trust and encourage repeat use, while frequent failures or frustrating interactions may hinder them from using this checkout method (Collier et al., 2017). Therefore, understanding customer loyalty in self-service environments is important to design systems that not only meet functional needs but also encourage long-term engagement and brand commitment.

Because of the strong link between service quality and customer loyalty, it is important to understand how service quality is measured. Several general approaches have been developed to assess service quality, which are applicable across various service settings. The next section reviews these methods for measuring service quality in retail environments.

2.1.3 Measuring retail service quality

Parasuraman et al. (1988) developed the SERVQUAL model, a widely utilized framework for evaluating service quality. This model continues to be a key tool in service quality assessment (Siu & Tak-Hing Cheung, 2001). SERVQUAL evaluates service across five key dimensions: tangibles, reliability, responsiveness, empathy, and assurance (Yuen & Chan, 2010). Later, Cronin Jr & Taylor (1992) proposed SERVPERF, a performance-only measure derived from SERVQUAL. SERVPERF had greater applicability and accuracy in assessing service quality. While SERVQUAL is widely recognized and utilized in various service environments, it is not applicable across different service contexts, especially in retail (Rao & Kelkar, 1997).

To better measure service quality in retail environments, Dabholkar et al. (1996) proposed the Retail Service Quality Scale (RSQS). This scale, based on qualitative research and existing literature, consists of five dimensions: (1) Physical Aspects - store appearance and layout convenience, (2) Reliability - fulfilment of promises and service accuracy, (3) Personal Interaction - staff courtesy, helpfulness, and ability to inspire trust, (4) Problem Solving - handling complaints and returns, and (5) Policy - store policies on payment, operating hours, and other customer conveniences (ibid). SERVQUAL and RSQS share similarities but differ in key areas. RSQS expands Physical Aspects to include store layout convenience. Reliability remains the same as SERVQUAL, while Personal Interaction combines Responsiveness and Assurance. RSQS also introduces two new dimensions: Problem Solving (handling issues) and Policy (merchandise quality) (Martinelli & Balboni, 2012).

Researchers have tested RSQS across various cultural and retail contexts. Siu & Tak-Hing Cheung (2001) re-evaluated it in Hong Kong, identifying a new dimension called "convenience". Mehta et al. (2000) compared RSQS in Singapore's supermarkets and electronics retailers and found better adaptability in supermarkets. Das et al. (2010) tested it in Kazakhstan but could not differentiate between store formats, similar to Boshoff & Terblanche (1997) in South Africa. Kim & Jin (2002) found differing perceptions between U.S. and Korean discount store shoppers. These studies highlight that service quality dimensions vary by industry, retail format, and culture. Given these findings, RSQS and SERVQUAL should be adapted before applying them across cultures (Meng et al., 2009)

This literature on service quality measurement is highly relevant to SCOs as it helps understand how customers perceive and evaluate service experiences. Models like SERVQUAL and SERVPERF emphasize reliability, responsiveness, and assurance that are key factors in ensuring seamless SCO

operations. The RSQS further highlights aspects such as physical layout, problem-solving, and policy, all of which influence SCO usability.

In the context of SCOs, physical aspects relate to kiosk design and ease of use, while reliability ensures error-free transactions. Though personal interaction is minimal, employee assistance remains crucial for issue resolution. Problem-solving focuses on resolving SCOs issues for customers. Meanwhile, policy factors, like available payment options and store hours, impact overall convenience of customers. These service quality frameworks can help understand customer expectations and identify areas where SCOs can be improved to enhance satisfaction and foster long-term customer loyalty.

2.2 Self-Service Technologies

2.2.1 Adoption of SSTs

SSTs refer to tools provided by organizations that allow customers to perform services on their own, without direct involvement from service staff (Hilton et al., 2013). This definition aligned with Meuter et al. (2000), who similarly describe SSTs as technological interfaces that enable customers to independently complete service tasks. According to (Dabholkar & Bagozzi, 2002), SSTs can be categorized into two types: on-site and off-site options. On-site SSTs include solutions like self-scanning in stores and libraries, touchscreen terminals at airports, and information kiosks at tourist centers. In contrast, off-site SSTs refer to services such as online banking and e-commerce (Considine & Cormican, 2016). Common platforms for self-service interactions include electronic kiosks, websites, phones, and mobile apps (Castro et al., 2010).

The applications of SSTs ranges from traditional solutions like ATMs to more advanced systems such as online banking, and self-service kiosks (Kelly et al., 2017). Self-service kiosks are considered as the most common application of SSTs (Cho & Fiorito, 2010). In the context of this study, self-service kiosks refer specifically to those used in retail and grocery environments (SCOs) that

allow customers to independently scan, bag, and pay for items. Self-service kiosks are interactive systems accessible to the public and capable of processing information (Meuter et al., 2000). Devices such as information kiosks, ATMs, vending machines, and similar technologies are categorized under the term "self-service kiosk." (Zahari et al., 2023). In the hospitality industry, kiosks are also used for automated hotel services, including self-checkin, room service ordering, messaging, and housekeeping requests (ibid). Emerging technologies like AI and AR are also being integrated to improve the customer experience with these self-service systems (Bigne, 2021).

SSTs were initially introduced to offer innovative service options and reduce labor costs, with cost reduction identified as the primary motivation for their adoption (Bitner et al., 2002). Beyond cost savings, SSTs provide several benefits, including improved operational efficiency, higher customer satisfaction and loyalty, standardized service delivery, and differentiation through technology (Meuter & Bitner, 1998). SSTs also enhance the service speed, customization, and accuracy of services (Berry, 1999), while contributing to productivity gains (Dabholkar et al., 1996). Moreover, integrating SSTs into service interactions helps organizations adjust their staffing levels according to fluctuations in customer demand (Curran & Meuter, 2005). By encouraging customers to perform certain service tasks themselves, providers can concentrate on priorities and reduce the need for staff to handle repetitive and routine duties (Castro et al., 2010).

2.2.2 Implementation of SCOs

Having explored the fundamentals of SSTs, the next step is to examine the specific solution of this thesis (SCO) to understand its function within today's grocery retail environment.

Self-checkout machines are automated systems that allow customers to scan, pack, and pay for their items independently (H.-J. Lee & Yang, 2013). SCOs

have made a significant and lasting impact on grocery store shopping. Rivera et al. (2021) analyzed 90 articles on the technological development within the grocery retail sector, emphasizing SCO as a key technology embraced by customers. SCO is an SST that plays a crucial role in enhancing the overall customer experience (Orel & Kara, 2014). Fernandes & Pedroso (2017) also highlighted that positive experiences with SCO can greatly increase customer loyalty, encouraging repeat visits.

Global adoption of SCOs is accelerating with projections indicating that installations will reach 2 million units worldwide by 2029 (Tatiana Walk-Morris, 2024). This growth reflects a big shift toward automation in retail.

The United States is the global leader in the adoption of SCOs with widespread implementation across grocery retail (Tatiana Walk-Morris, 2024). In 2024, the United States accounts for over 40% of the global SCO market with 325,000 units (Charles Scott, 2024). This widespread implementation is due to major supermarket chains and convenience stores that aim to enhance efficiency and meet consumer preferences (Capital One Shopping, 2025).

Europe accounts for 25% market share of SCO globally (Charles Scott, 2024). The United Kingdom leads in SCO adoption with approximately 105,000 units currently active across the country (Charles Scott, 2024). Germany and France also demonstrate strong usage, with SCOs available in over 61% of supermarkets (Global Growth Insights, 2025).

Across the Asia-Pacific region, SCO adoption is accelerating rapidly. China is the leader of the region with over 74% of retailers in tier-one cities implementing mobile-integrated checkout kiosks (Global Growth Insights, 2025). Japan is not far behind, with more than 63% of convenience store transactions automated through SCOs (Global Growth Insights, 2025).

While comprehensive market statistics for SCO adoption in Finland remain limited, consumer acceptance of the technology is notably high. According to a study by the Finnish Commerce Federation with feedback from 8,000 customers, 90% of respondents found self-service shopping to be easy and accessible (Yle, 2017). This strong consumer approval is reflected in usage figures: in 2017, K Group reported that around 20% of its customers use SCOs weekly, translating to approximately 400,000 users (Yle, 2017). In Finland, both leading grocery chains, S Group and K Group, provide SCOs as a payment option for customers (Elina Kaakinen, 2021). S Group was the first to implement SCOs on a large scale in 2012 (Tapani Rinta-Kahila, 2013). S Group had an earlier unsuccessful attempt in the 1990s (ibid). A similar pattern was seen in the U.S., where early efforts to introduce SCOs also faced challenges (Dabholkar & Bagozzi, 2002). This highlights the importance of timing in promoting new technology adoption, which depends not only on consumer readiness but also on the competitive landscape (Audrin, 2020).

2.2.3 Impacts on customer behavior

Even though a company may gain short-term benefits from SSTs, forcing customers to use them can create problems (Plé & Chumpitaz Cáceres, 2010). Therefore, businesses should understand how willing their customers are to use these technologies. Hilton et al. (2013) point out that people's knowledge and skills with technology affect which SSTs they use, while Meuter et al. (2005) say that trials of these new technologies are also important for customers. Similarly, Liljander et al. (2006) and Meuter et al. (2005) explain that a customer's attitude and their ability and willingness to complete tasks influence how they feel about SSTs.

The Technology Acceptance Model (TAM), developed by Davis (1989), suggests that perceived usefulness and perceived ease of use influence whether users adopt a new technology. This framework has guided much of the research on SSTs. How customers perceive a technology's ease of use influence their acceptance of it (Chang & Chen, 2021), especially for SSTs

(Venkatesh & Bala, 2008). Meuter et al. (2005) emphasize that the ease of using and learning is important for adopting SSTs. Dean (2008) shows that older generations tend to trust SST transactions less because they may struggle with technological interfaces. Venkatesh et al. (2003) also highlight how age affects ease of use and technology adoption. Liljander et al. (2006) explore technical readiness in SST acceptance. They find that there are different levels of readiness for users and non-users, which depends on their technology skills. As Galdolage (2021) notes, detailed information and step-by-step guidance on many SST platforms help customers learn how to perform self-service transactions. While TAM focuses on individual perceptions, Orlikowski (1992) offers a broader sociotechnical perspective: technology use is shaped not only by user attitudes but also by organizational routines, social norms, and contextual factors. This view is relevant for this study's setting since both customers' experiences and employees' actions influence how SCOs are used and accepted.

Additionally, as Galdolage (2020) pointed out, learning to use SCOs is mostly a self-guided process. Therefore, a customer's willingness and ability to learn how to use SSTs can increase their preference for using SSTs over traditional service options (Chang & Chen, 2021). Research has shown that users' anxiety about technology is often due to a lack of technical knowledge, which negatively impacts their use and evaluation of SSTs (Meuter et al., 2005). Aimin et al. (2016) also explain that anxiety and a lack of trust in technology contribute to dissatisfaction and reduce the willingness to use SSTs.

Customer discomfort with technology, lack of human interaction, concerns about service failure, and employee dissatisfaction are often drawbacks of SSTs (Curran & Meuter, 2005). Employee support plays a crucial role in the effective use of SCOs in supermarkets (Anitsal & Paige, 2006). In other words, effective human coordination is key to the smooth functioning of retail kiosks (Rowley, 1995). Well-trained employees can offer quick and

efficient help when customers encounter issues with kiosks, which helps minimize the impacts of service failure (Cho & Fiorito, 2010).

To enhance customer satisfaction and loyalty, service providers must adopt the latest SST designs and technologies in line with consumer preferences (Parasuraman et al., 2005; Tam, 2023). The design of a kiosk's user interface plays a crucial role in helping customers navigate its functions, therefore affecting its overall efficiency and effectiveness (S. Galdolage, 2020). Although kiosks are designed for consumers' usage, employee assistance and training are still necessary to improve customer satisfaction and prevent system failures (Porter & Gibbs, 2001).

This section 2.2 emphasizes the critical factors that influence customer behavior when using SCOs. It highlights that customer willingness to use SCOs depends on their technological skills, familiarity, and comfort with the interface, as well as their ability to learn how to operate the system. Perceptions of ease and trust, especially among older users, are crucial for adoption. Effective design and user interfaces are important in facilitating smooth interactions because customers are more likely to engage with systems that are intuitive and easy to navigate. Additionally, employee support plays a vital role in reducing service failures and ensuring customer satisfaction. Overall, understanding these factors and designing systems that align with customer preferences can enhance the customer experience, improve adoption rates, and reduce dissatisfaction.

2.3 SST Recovery

2.3.1 Service recovery definition and literature summary

Service failure occurs when the provided service does not meet the customer's expectations during their interaction with the service (Chou, 2015). Similar to traditional services, failures in SSTs are also common - even when no actual technical issue occurs - which results in customer dissatisfaction (B. Lee & Cranage, 2018; Meuter et al., 2000). Therefore,

addressing these SST failures quickly, referred as service recovery, is a valuable chance for service providers to restore customer satisfaction (Dabholkar & Spaid, 2012), enhance perceptions of fairness (Mattila & Cho, 2011), and increase the likelihood of future purchases (Roggeveen et al., 2012).

Strategies for recovering from SST failures can be based on the extensive research on traditional service recovery (Forbes, 2008). However, in SST settings, technology plays a more significant role than employees during service encounters. Common issues in SSTs are different in nature, compared to issues in traditional service contexts (Kelley et al., 1993; Meuter et al., 2000). Additionally, from a consumer point of view, the time-saving advantage of SSTs compared to other service options must be considered when addressing failures (Dabholkar & Spaid, 2012; Meuter et al., 2000). However, research on SST recovery remains limited, as outlined in Table 1 below.

Study	Objectives	Methodology	Findings
Forbes, 2008	To explore customer responses to SST failure and recovery in non-internet settings.	Experimental design with scenarios involving SST failures.	Customers prefer quick, efficient resolution; organization-initiated recovery leads to better perceptions of justice
Mattila & Cho, 2011	To examine how different types of justice (distributive, procedural, interactional) influence satisfaction in SST failure recovery.	Scenario-based experiment with manipulated justice elements.	Interactional justice (e.g., empathy) still matters even in SST contexts; perceived fairness boosts satisfaction.

Dabholkar & Spaid, 2012	To study the effects of failure type and recovery method on satisfaction and attribution	Experimental design using SST scenarios with varying failure types and responses.	Fast recovery is crucial in SST; customer satisfaction is influenced by perceived controllability and responsibility.
Zhu et al., 2013	To investigate how customers recover from SST failures on their own and what affects their loyalty.	Field survey and lab experiment.	Customers who succeed in self-recovery are more likely to stay; perceived effort and success drive decisions.
Collier et al., 2017	To examine customer preferences in SST recovery, focusing on social context and self-monitoring.	Lab experiments testing social presence and behavior monitoring effects.	Presence of others affects preferences; self-monitoring can increase tolerance for employee involvement.
Dao & Theotokis, 2021	To study how the initiator of recovery (customer vs. employee vs. system) and responsibility attribution impact repurchase intent.	Online experiments using manipulated scenarios.	SST-initiated recovery is often more effective; responsibility attribution moderates satisfaction and loyalty.
Chiu & Nguyen, 2022	To explore how customers' autonomy needs affect their	Survey and structural equation modeling based	Autonomy and competence significantly influence

	willingness to engage in self-recovery.	on Self-Determination Theory.	customers' motivation to recover independently.
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Table 1. Overview of SST recovery studies

2.3.2 Customer role in service recovery

In service encounters, customers play an important role in the service delivery to co-create value (Roggeveen et al., 2012). Based on this perspective, customers should also be actively involved in the recovery process to help enhance service quality and customer satisfaction (Dong et al., 2015; Xu et al., 2014). Customer participation in service recovery has gained growing interest, particularly in technology-driven services (Collier et al., 2017; Dao & Theotokis, 2021; Xu et al., 2014; Zhu et al., 2013). Collier et al. (2017) show that social context and self-monitoring affect preferences for employee help in SST failures. Dao & Theotokis (2021) emphasize the importance of who initiates the recovery process and who is perceived as responsible for the failure. Meanwhile, Zhu et al. (2013) focus on self-recovery, analyzing how customers choose to resolve issues independently and what drives their decision to continue or abandon SST use. Xu et al. (2014) found that employee-initiated recovery boosts satisfaction and repurchase intent, but only when there is collaboration between the customer and the company during the recovery.

Customer participation in SST recovery is shaped by various factors, including responsibility attribution, social context, and personal agency. This participation can vary to different extents, ranging from self-recovery (where the customer resolves the issue independently without employee involvement), to joint-recovery (where the customer collaborates with employee for a solution), and firm-recovery (where the customer remains passive and the service provider handles the entire recovery process) (Chiu & Nguyen, 2022).

2.3.3 Recovery initiation

Previous studies on SST recovery have primarily concentrated on customer-initiated efforts, such as complaints, customer-solving problems, and individual failure-handling strategies, for example studies of Collier et al., (2017; Mattila & Cho (2011); Zhu et al. (2013). Another key approach is organization-initiated recovery, where the service provider proactively identifies and addresses service failures before customers raise any issues (Smith et al., 1999). This proactive method has been shown to improve customer evaluations after recovery (Patterson et al., 2006; Smith et al., 1999) and is increasingly relevant in SST environments. In SST contexts, organizational recovery can be either by an employee (employee-initiated) or automatically by the system itself (SST-initiated). This raises important managerial and theoretical questions about the conditions under which these proactive approaches may be more effective than customer-initiated recovery.

In traditional service contexts, companies often take the lead in initiating recovery efforts (organization-initiated recovery) rather than waiting for customers to file complaints (Smith et al., 1999). Research has shown that customers tend to favor this proactive approach over customer-initiated recovery (Kelley et al., 1993; Smith et al., 1999), suggesting a similar preference might apply in SST settings. However, the dynamics differ in self-service environments. One key reason customers choose SSTs is to avoid human interaction (Meuter et al., 2000), which may be contradicted when service recovery is handled by employees (Mattila & Cho, 2011). While organization-initiated recovery might be less effective in SSTs for this reason, it can still offer value through quick resolutions (Dabholkar & Spaid, 2012). SST environments offer a unique opportunity for the technology itself to initiate recovery. However, empirical research exploring the effectiveness of such SST-initiated recovery remains lacking. Moreover, the literature

remains inconclusive on whether organization-initiated recovery outperforms customer-initiated recovery in SST settings.

2.3.4 SST recovery summary

Based on the reviewed literature, it is evident that SST recovery is a complicated process shaped by both technological and psychological factors. While traditional service recovery insights remain relevant, the unique nature of SSTs with their lack of human interaction and customers' desire for autonomy requires adapted recovery strategies. Studies consistently highlight that quick, seamless, and appropriately initiated recovery (whether by the customer, employee, or the system) plays a critical role in restoring satisfaction.

Moreover, customer preferences vary depending on the failure context, social setting, and perceived responsibility. While some customers favor independent resolution (self-recovery), others appreciating proactive, system- or employee-led interventions. Overall, successful SST recovery depends not only on technological responsiveness but also on understanding customer psychology - particularly their need for control, competence, and convenience during the recovery process.

These insights are very relevant to SCOs, where service failures such as scanning errors, payment issues, or system freezes can quickly lead to customer frustration. Given the minimal staff interaction and high expectations for speed and autonomy, effective recovery in SCOs must be fast, intuitive, and tailored to user needs. Since recovery can be initiated by the system, by nearby employees, or handled by the customer, aligning the approach with users' desire for control and convenience is important to maintain satisfaction and encourage continued use of SCOs.

2.4 Customers' attitudes towards SST recovery

2.4.1 Environmental anxiety and customer perceptions

In-store anxiety caused by crowding or the presence of other shoppers can negatively impact customers service experiences with SSTs (Dabholkar & Bagozzi, 2002). However, this type of anxiety has not been explored in SST failures and recovery efforts. It is still reasonable to assume that such environmental stress could influence how customers behave in these settings.

People's interpretation of events can be influenced by their sense of control when lower perceived control tends to result in more negative attitudes (Pittman & Pittman, 1980). When interacting with SST, customers may feel a loss of control during a service failure. Environmental factors such as kiosk placement, store crowdedness, noise, or lack of privacy can intensify this feeling (Dabholkar & Spaid, 2012). On the contrary, when the surrounding environment is calm and non-stressful, users are less likely to feel overwhelmed (ibid).

Chiu & Nguyen (2022) highlight how customers' recovery experiences in SSTs are impacted by their perceived competence, autonomy, and relatedness. Customers' perceived competence refers to their belief in their own ability to successfully carry out the recovery process on their own (Ryan & Deci, 2020). When customers manage to fix SST failures by themselves, they may experience benefits such as feeling accomplished, gaining new knowledge or skills, or enjoying the process (Dong et al., 2008; Yim et al., 2012). Perceived autonomy is defined as the customer's feeling of having the freedom to decide how to handle the recovery process (Ryan & Deci, 2020). When encountering failure, customers can choose self-recovery over alternatives like staff assistance or joint recovery, which is a trade-off because it requires more personal time, effort, and cognitive involvement (Zhu et al., 2013). Perceived relatedness is defined as the customer's expectation that the service provider recognizes and values their efforts in resolving the issue (Ryan & Deci, 2020). When an SST failure happens, customers voluntarily resolve the error proactively themselves. They may expect the service

provider to acknowledge their actions, which strengthens the customer-provider relationship (Chiu & Nguyen, 2022).

When customers feel competent, autonomous, and supported by the service provider, they are more likely to engage in self-recovery and feel satisfied with the outcome. However, a stressful environment can undermine these perceptions, making customers feel less capable, less in control, and more isolated, which in turn reduces their willingness to resolve the issue independently.

2.4.2 Source of failure

SST failures generally fall into two categories: technology-related and customer-induced (Dabholkar & Spaid, 2012; Dao & Theotokis, 2021; Meuter et al., 2000). It is important to understand how each type influences customer attributions and satisfaction. As STTs become more advanced, customers often expect them to function smoothly, so a system malfunction can come as a surprise, which make customers attribute negative attitudes toward the kiosk and the store (Dabholkar & Spaid, 2012). However, prior research also shows that customers acknowledge their own role in some failures and are often willing to share the blame (Harris et al., 2006; Meuter et al., 2000). As a result, when the issue comes from a customer error, they are likely to view the system or service provider less negatively compared to when the failure is caused by a technical malfunction (Dabholkar & Spaid, 2012).

2.4.3 Employee assistance

Research on SST failures has explored various recovery strategies, but customer perceptions of employee assistance remain complex and underexamined. Dabholkar & Spaid (2012) found that while employee involvement can enhance perceived control during a service failure, this support may shift customers' negative attributions from the employee to the technology itself, especially when the technology fails to deliver the promised

self-service interactions. Similarly, Mattila & Cho (2011) showed that compensation offered by employees might be more effective in restoring justice with traditional failures (matched condition) than with SST failures (mismatched condition). Collier et al. (2017) emphasized that customers value justice and control in failure recovery. Therefore, when employees assist quickly and appropriately, it can restore fairness and autonomy (ibid). However, H.-J. Lee (2015) highlighted that while some users appreciate employee support during SST issues, others think it undermines the efficiency and independence promised by SSTs.

The discussion on SST recovery is relevant to SCOs, as environmental factors can reduce customers' sense of control during failures, making them less likely to handle issues themselves (Chiu & Nguyen, 2022). While employee assistance can restore fairness, it may also disrupt the independence SCOs promise (Dabholkar & Spaid, 2012). Understanding these dynamics is crucial for improving SCO recovery strategies and customer experience.

However, prior literature has paid limited attention to the consequences of such assistance for employees themselves. Drawing from emotional labor theory (Hochschild, 2002), this study reveals that supporting SCOs often involves 'invisible' work. It can be defined as tasks that are not formally acknowledged or compensated, such as managing customer frustration, masking one's own emotions, and continuously monitoring multiple kiosks. These demands require emotional control and multitasking that exceed physical troubleshooting.

This aligns with (Giebelhausen et al., 2014), who argue that employees often perform unrecognized emotional work to uphold service experiences, especially during failures. Our findings further extend this perspective to SCO contexts, where employees are required to ensure the success of a technology that is expected to be self-sufficient.

Therefore, this study contributes to the research gap by reframing SCO recovery not only as a technical or customer experience issue but as a form of redistributed labor. It shows that while SCOs seem efficient, they often hide extra work for employees such as managing emotions, thinking quickly, and handling interruptions. This perspective has been underexplored in SST literature, which tends to prioritize user satisfaction and technical performance over employee well-being and role complexity.

3 Research methodologies

This chapter outlines the research methodologies and empirical process used in this thesis. It begins with an overview of the research design. The research setting and sample are then described, focusing on stores from S Group and K Group. The data collection process is explained, including how participants were selected and how the interviews were conducted. This is followed by a description of the data analysis approach.

3.1 Research design

The research design of this thesis follows a design science approach. This approach is particularly suitable for addressing real-world problems by combining practical relevance with theoretical contributions.

Design Science aims to produce knowledge that is both practical and scientifically relevant (Holmström et al., 2009). Unlike traditional research approaches that primarily try to explain or understand existing phenomena, Design Science focuses on the creation of models, methods, or systems to address complex, real-world problems (March & Smith, 1995). It is a problem-solving approach that bridges the gap between theoretical research and practical application. In contrast to traditional research approaches that focus mainly on understanding phenomena, design science emphasizes the creation of innovative solutions to real-world challenges. As described by (Hevner et al., 2004), design science contributes to the development of technology and practice while ensuring that the outcomes of the research are grounded in theory and can be generalized to other settings. Furthermore, Gregor & Hevner (2013) emphasize the importance of iterative cycles of building, evaluating, and refining solutions. This makes Design Science particularly suitable for information systems and service innovation contexts where system design and user experience are focused.

This approach is relevant for several reasons. First, the primary objective of design science is practical problem solving since it aims to develop solutions for real-world issues (the challenges associated with SCOs in retail settings). Second, while addressing these practical concerns, design science also supports theory development by contributing frameworks or models that explain how specific designs can enhance system performance or user experience (Hevner et al., 2004).

This methodology allows the focus on improving SCO systems in Finnish grocery retail stores by designing solutions to current issues identified from employee experiences. With qualitative semi-structured interviews with employees from S Group and K Group, valuable insights are gathered into the operational challenges with SCO systems, which can be used to inform improve systems or processes.

3.2 Research setting

To understand the variety of stores and the scale of the Finnish grocery retail sector, some figures are important to mention. In 2024, the total turnover of the Finnish grocery store industry was €23.5 billion. S Group and K Group are the two dominant players with 48.8% and 33.7% of the market share respectively (Finnish Grocery Trade Association, n.d.). This translates to a combined turnover of approximately €19 billion. Given this oligopolistic market structure, the study focuses exclusively on these two groups. Their widespread presence and influence make them highly relevant for examining SCO systems since their operational decisions significantly impact the majority of Finnish consumers. By narrowing the scope to S Group and K Group, the research ensures that findings are both meaningful and representative of the broader industry.

Table 2 provides an overview of the main store chains operated by S Group and K Group. Since the thesis focuses on SCO experiences and challenges, understanding these store formats is crucial. Each store type varies in terms

of size, customer flow, service design, and technology deployment. These characteristics can influence how SCO's are implemented and perceived by employees and customers. Only Prisma, S-market, Sale/Alepa (S Group) and K-Citymarket, K-Supermarket, K-Market (K Group) are included in this study. Employees from these stores were interviewed to capture a diverse range of experiences with SSTs across different store formats.

Store Type	S Group	K Group	Description
Hypermarket	Prisma	K-Citymarket	Large-format stores offering groceries, clothing, electronics, and home goods
Supermarket	S-market	K-Supermarket	Mid-sized grocery stores with a wide selection of everyday and specialty food items
Convenience store	Sale / Alepa	K-Market	Smaller grocery stores for daily needs and convenience shopping
Premium grocery	Food Market Herkku	Some premium K-Citymarkets	Focus on high-end groceries and specialty products
Service station	Part of ABC chain (includes ABC!)	Neste K	Combines fuel station with a convenience store and ready-to-eat food

Table 2. Overview of Major Store Chains Operated by S Group and K Group
(Finnish Grocery Trade Association, n.d.; Kesko, n.d.; S-Ryhmä, n.d.)

3.3 Data collection

The selection criteria for the interview sample in this study were guided by the need for relevant and knowledgeable participants (Patton, 2002). The sample of employees from S Group and K Group stores is narrowed down to those who meet specific criteria to ensure the participants can meaningfully contribute to the research.

The inclusion criteria for interviewees are as follows:

- **Current Employment and Experience:** Participants must still be working at one of the selected stores (Prisma, S-market, Alepa from S Group; K-Citymarket, K-Supermarket, K-Market from K Group) and have been employed for at least one year. This ensures that interviewees have sufficient experience with SCOs.
- **Selection via LinkedIn:** Potential interviewees are contacted through LinkedIn, where their job roles are identified. They are informed about the research requirements, so that transparency and clarity on the expectations were guaranteed.

These criteria ensure that interviewees have the expertise and direct experience with SCOs. Only employees who meet these requirements can be included in the sample. This study includes 10 interviewees, which aligns with the recommended range for interview-based research. Robinson (2014) notes that qualitative studies typically involve between three and sixteen participants, depending on the research scope and depth of inquiry. Rather than aiming for a specific number, this study prioritized information richness and relevance when selecting participants.

Data analysis was conducted alongside the interviews to assess when insight saturation had been reached, which is when additional interviews were unlikely to provide new insights (Strauss & Corbin, 1998). By the 8th and 9th

interviews, the findings began to show strong repetition in themes, with no significant new insights emerging. A 10th interview was conducted to confirm this observation and ensure robustness. As no new substantial themes were identified, the sample of 10 interviews was considered sufficient to support the research objectives. Table 3 presents the anonymized list of interviewees along with background details they voluntarily shared.

Interviewee	Store group	Store type	Years of experience	Length of Interview (minutes)
1	S Group	Convenience store	2	45
2	K Group	Convenience store	2	50
3	K Group	Convenience store	3	40
4	K Group	Hypermarket	1	50
5	S Group	Hypermarket	2	60
6	K Group	Convenience store	2	70
7	K Group	Supermarket	4	60
8	S Group	Supermarket	2	30
9	S Group	Hypermarket	3	45
10	S Group	Convenience store	1	30

Table 3. Interviewee profile

Data was collected, with participants' consent, through 10 semi-structured interviews conducted via Zoom. All interviews were in English, recorded, and transcribed with interviewees' permission. A question list (see Appendix 1) guided the discussions, with follow-up questions and personal examples. Due to the small pool, no demographic diversity criteria were applied. All transcripts were securely stored on the researcher's personal laptop with access limited to the researcher. All raw data was anonymized before analysis to protect interviewee privacy and minimize researcher bias. After the completion of the study, the data was securely destroyed to comply with ethical research guidelines.

3.4 Data analysis

Repeated observations are essential in qualitative research to ensure the validity and reliability of findings (Terry et al., 2017). During each interview, handwritten notes were taken to record keywords, recurring themes, and memorable statements. These notes were reviewed immediately after each session to ensure data quality and to identify emerging themes that could inform and adjust the direction of subsequent interviews. This iterative process allowed the interview questions to remain responsive to insights developing during the study (Gioia et al., 2013)

After completing all ten interviews, the recordings were reviewed in full, and key discussion points for each participant were extracted and documented in a Google Sheet; this is inspired by the Gioia Method. Transcripts were analyzed individually, and the data was synthesized using a two-step coding process. In the first column of the Sheet file, relevant quotes and raw responses were recorded; in the second column, thematic codes were assigned to each segment based on content. This structured approach helps maintain rigor and transparency while enabling the emergence of data-driven, grounded theory (Gioia et al., 2013; Nowell et al., 2017).

The data analysis was guided by the thematic analysis approach of Terry et al. (2017), which offers a structured but flexible framework for identifying, analyzing, and reporting patterns for qualitative data. The process followed Clarke's six-phase method (Braun & Clarke, 2006): (1) familiarization with the data, (2) generating initial codes, (3) searching for themes, (4) reviewing themes, (5) defining and naming themes, and (6) documenting data analysis. This approach was suitable to the exploratory nature of the study because it does not require pre-existing coding frames and allows themes to emerge from the data.

4 Findings

This chapter presents the findings identified from employees' answers. The analysis reflects employee perspectives on common SCO issues, how users respond to them, and the role employees play in resolving these problems. It also captures employees' perspectives on balancing efficiency with customer support and offers practical suggestions for system and process improvements.

4.1 SCO Errors: User Autonomy & Employee Assistance

This section groups the findings into key themes that represent common problems encountered by customers at SCOs. Each subsection describes a typical scenario, identifies what goes wrong and highlights the types of interventions required from employees. These insights reveal the limits of customer autonomy in resolving these common errors and illustrate how employee support becomes essential in ensuring smooth SCO operations. Direct quotes from interviewees are used to illustrate real-world examples and emphasize the frequency and impact of these issues.

4.1.1 Item Quantities and Scanning Mistakes

A frequent and recurring challenge at SCOs is related to item quantity errors, especially when customers accidentally scan the same product multiple times. In current SSC systems, customers cannot resolve these mistakes themselves, which often leads to unnecessary employee intervention. This issue was highlighted by all interviewees, suggesting that it is one of the most common user-side errors in daily operations.

"The most common issues are related to customer mistakes, for example, scanning the same product twice. Being able to remove a product that you have accidentally scanned twice would make interventions by the cashier much less common." (Interviewee #1)

While enabling customers to remove mistakenly scanned items seems like a logical improvement, several employees raised concerns about the potential misuse. Allowing quantity adjustments could create opportunities for dishonest behavior, such as deleting items after appearing to have scanned them.

"To know someone is stealing, I listen to the scanning sound. If they scan five items, I hear five beeps. If I hear only four, I go over. If customers could delete items themselves without any feedback, it would be easier to steal." (Interviewee #4)

To mitigate this risk, many interviewees proposed a sound-based alert system to signify quantity changes. A unique, loud sound could signal that a quantity was edited, helping employees distinguish legit adjustments from potential misuse.

"It can be a different sound from the scanning sound. I think that's a really good idea. As a cashier, I would appreciate that." (Interviewee #8)

"It would make our jobs easier if they [the customers] could remove them [accidentally scanned items] themselves, but there should be protocols to decrease the scammers for trying to scam." (Interviewee #5)

The balance between customer convenience and theft prevention is a dilemma. While several employees supported giving users more control, most agreed this should be combined with visibility and accountability tools for employees.

Some stores have started addressing this issue with remote monitoring capabilities that allows employees to view and manage SCO transactions from their cashier stations. This system enables employees to correct errors

such as accidental double scans, approve age-restricted items, and extend timeouts without physically approaching the machine. While this technology has the potential to streamline many other types of interventions, its current implementation is limited to smaller stores where visibility and employee multitasking are more manageable. In larger stores, where dedicated SCO assistants are often present, the use of such remote systems is less common and may be considered less necessary. This new system is being piloted in a K-Market:

"Kesko [K Group] has recently introduced a beta version of a system that allows us to control all self-checkouts remotely from our cashier station through a monitor. We can see all scanned items and even approve certain actions without physically going to the self-checkout area." [Interviewee #2]

This approach reflects an effort to improve customer autonomy while maintaining supervision. However, it still requires employee judgment and cannot completely replace in-person support in more complex situations.

In summary, quantity-related errors are one of the most common and frustrating issues at SCOs. While employees generally support more user-friendly systems, they also emphasize the need for close supervision.

4.1.2 Alcoholic and Age-Restricted Products

Purchasing alcoholic beverages and other age-restricted items at SCOs requires employee assistance. These interruptions are necessary for legal reasons, but they also cause disruptions in the customer purchasing process and increase employee workload, especially during busy hours.

The standard procedure is that the kiosk stops the transaction and notifies an employee to verify the customer's age. Interviewees noted that this

process applies not only to alcoholic beverages but also to related products such as 0% beer, fermented drinks like Sima, and even energy drinks.

“Even though it's beer with 0%, it still has to [be checked] ... and then also Sima, because apparently with the fermentation process, it can get a little bit of alcohol... energy drinks also stop the process and ask for intervention.” (Interviewee #3)

While this check is required in every store, the way it is implemented varies across stores. Some employees can approve the purchase remotely, while others still need to physically go to the kiosk. Interviewee 6 shared how their K-Market uses a remote interface with predefined responses:

“There's like three buttons... the green one is no need for ID check, continue. The yellow one is check the ID and allow. And the red one is don't sell, and then you have to choose the reason.” (Interviewee #6)

This remote capability offers flexibility and saves time when staff can confidently assess the customer with an clearly overage.

“They can pay [for age-restricted items] at the self-checkout point, but the checkout point always asks for an employee to check the ID if needed.” (Interviewee #9)

In summary, SCOs should be equipped to handle legal requirements around age-restricted products since the interaction often disrupts flow for both customers and employees. While some stores have implemented remote approval systems, others still rely on manual verification. A practical solution to reduce staff interventions for age-restricted purchases is to integrate digital ID verification directly into the self-checkout process. Customers could verify their age using secure government-issued digital IDs in apps like Apple Wallet or Finland's digital driving license (Traficom, 2025). This technology enables fast, contactless confirmation via QR code or NFC (ibid).

This approach would maintain legal compliance while improving the speed and autonomy of SCO transactions.

4.1.3 Timeout Notifications

Timeout errors at SCOs are a frequent problem that causes confusion and requires employee intervention. The error occurs when the system detects inactivity, which is often between scanning items and completing payment. Therefore, the machine automatically assumes that the user has abandoned the transaction. While intended as a security feature, the timeout function frequently interrupts purchasing process, especially among slower or older users. Although the exact timeframe is not identified for every store, all interviewees agreed that it is too short, which causes many unnecessary interventions.

“Timeframe [of] 45 seconds is too short and not enough time for customers to find payment card and bonus cards.” [Interview #7]

“Our customer base is a bit older, so they tend to move a bit slower... and then the machine goes into timeout mode. Customers get frustrated because they feel like they’re being punished for being a bit slow.” [Interviewee #4]

To mitigate these disruptions, many stores have adopted remote control systems that allow staff to extend the session without physically approaching the kiosk.

“We can now clear this alert remotely from our cashier station, so we don’t always need to go to the self-checkout area to resolve it.” [Interviewee #8]

Despite this advancement, the underlying issue remains. Customers cannot manage the timeout themselves. Several interviewees proposed a simple “I need more time” button that customers could press to extend their session.

“It would be beneficial to extend the timeout... A button [that lets them say] ‘I’m still here’ would be a good idea.” [Interviewee #4]

“Customers just need a bit more time to find their card. It would help a lot if they could press something themselves.” [Interviewee #10]

In summary, the timeout feature often misinterprets slow behavior as abandonment, which directly affects older or distracted customers. While remote intervention has improved employee response, enabling users to manage their own session with a visible "More Time" button or longer default timeout durations could further reduce disruptions. This solution is not only simple and low-cost to implement, but also scalable across different store types.

4.1.4 Bottle Deposit Refund

In Finland, the pantti system refers to the bottle deposit refund program, where customers receive a receipt (panttikuitti) after returning recyclable bottles or cans through a reverse vending machine (Palpa, n.d.). This receipt can be used as a form of payment or discount toward purchases (ibid). However, at SCOs, the handling of pantti receipts remains a persistent usability issue. Customers cannot use the pantti receipts at SCOs independently since the system does not recognize if they scan the receipts multiple times. Therefore, employees need to scan and take away the receipts to prevent duplicate attempts.

“You can’t use pantti at self-service checkouts. It needs staff there to scan the pantti receipt.” [Interviewee #6]

This limitation confuses many users, particularly those who are unfamiliar with the system.

“The small ticket from recycled bottles - they can only use with the cashier. The people who are not from Finland don’t even know about the bottle system.” [Interviewee #3]

The design and user interface of SCOs often fail to clearly communicate the restriction on pantti usage. Interviewee 10 suggested that the issue stems from poor communication. There should be clear signs at the front of the SCOs explicitly stating that pantti receipts cannot be used there. This can prevent confusion before customers begin the checkout process.

“I feel that most problems with pantti kuitti could be designed better through a more informative sign. For now, the self-checkouts have information about these things, but they are clearly not simple or informative enough since these problems arise regularly.” [Interviewee 10]

A technical solution to the pantti issue can be barcode validation and user interface enhancements at SCOs. Each pantti receipt includes a unique barcode for automatic recognition and validation. This would ensure that the value is applied only once and prevent repeated scanning of the same receipt. Barcode-based validation is already widely used in retail to ensure accuracy and prevent duplication in product scanning (ConnectPOS, 2023). Moreover, clear signs near SCOs should indicate whether pantti receipts can be used at SCOs and how to use them. This multi-layer approach would reduce unnecessary staff intervention, increase system security, and improve customer experience.

4.1.5 Discounted Items or Coupons

Discounted items, especially those nearing expiry, are a common source of confusion and error at SCOs. In Finland, such products often receive manual discount stickers (e.g. -30% or -50%) (Jorg Snoeck, 2019). However, store policies, labeling practices, and system capabilities vary across stores and

between S Group and K Group, which leads to inconsistent customer experiences and frequent staff intervention.

Some stores have optimized the process by using discount stickers with unique EAN barcodes. When scanned, the SCO system automatically applies the correct reduced price.

“Our store now uses a printed discount label for discounted products with a new EAN code. This allows customers to scan the discounted product at the self-checkout, where the system automatically recognizes the discount.” [Interviewee #2]

However, many stores - especially smaller ones - do not use barcoded discount labels or only use them selectively.

“We don’t have barcode for discounted items such as minus 30%. Their prices don’t go automatically.” [Interviewee #5]

“Our small store doesn’t have those [barcode for discounted items] since the store owner prefers it goes through the cashier.” [Interviewee #3]

“A product with a discount label needs to be scanned at the normal check-out counter. Products that are on sale can’t be sold at the SCOs.” [Interviewee #10]

Even with a barcoded discount, customers can accidentally scan the original product barcode, causing a price mismatch. This mistake causes confusion when the displayed price does not reflect the expected discount for customers.

“Discounted product with barcode that are stucked at the wrong place [not directly onto the original barcode] can cause problem when

customers accidentally scan the original barcode with the original price.” [Interviewee #7]

“If they accidentally read the old label and not the new one we printed... then it’s not 30% off.” [Interviewee #6]

The issue becomes even more complex when a product is part of a campaign promotion for example ‘buy 6 for €2’. Sometimes the campaign is implemented in the SCO system, and sometimes it is not, so the purchase might need a manual discount. In these cases, customers notice a pricing conflict and then require staff to manually enter the correct price.

Some stores attempt to communicate these limitations through checkout signs, but instructions are often text-heavy or unclear.

“We have a sign about this on the checkout entrance, but some customers don’t read it. I think it would be better if the instructions had pictures, like if it has this kind of sticker, call the cashier, but if it has this kind of sticker, you can scan it yourself.” [Interviewee #4]

While the underlying discount system often lies outside the SCO design, employees agree that both technical and communication improvements are needed.

“Creating a different kind of system for putting products on sale could help, but this would be out of the scope of SSC design and more in the realm of other systems.” [Interviewee #1]

4.1.6 Payment Interface Friction

Payment-related issues at SCOs often come from unclear instructions, system features or interface delays, especially when using loyalty cards, gift cards, or attempting a retry after a failed payment. One frequent source of disruption is the need to manually press the "Proceed to Pay" button even

after scanning a loyalty card like the K-Plussa card. Interviewee 6, who works at a K-Market, observed:

“Customers try to read their card, but they don’t realize that they have to press the button called ‘Proceed to Pay’ or else the machine stands still. If they don’t have the Plussa card, there’s also ‘I don’t have a Plussa card.’ Button. You have to press that.” [Interviewee #6]

This extra step slows down the process and often leads to confusion, particularly when users assume the scanned loyalty card should automatically advance the transaction. This step confuses customers, resulting in a timeout error if the system detects inactivity. This error is discussed above. This error requires employee help and interrupts the purchasing process. The payment step is implemented better in S-Group stores.

“If you show your S-Card in S-Market, it doesn’t ask to proceed to pay again. It just goes to payment straight.” [Interviewee #6]

When it comes to payment failures, whether from card issues or insufficient funds, the current system in K-Group stores typically requires staff intervention to allow a retry.

“Payment failure at self-service checkouts are usually caused by the machine not reading the card or a card being declined. In most cases, the cashier just instructs the customer to try again, which resolves the issue 90% of the time. Currently, the cashier must manually log in to allow a retry, but it would be better if the system allowed customers to retry payment on their own without employee assistance.” [Interviewee #4]

Another common pain point involves gift card compatibility. While the store’s own gift cards usually work seamlessly, third-party or shopping mall

gift cards often require a customer signature, which prompts a call for assistance.

“Shopping mall gift cards always require a signature, which triggers employee assistance. But our own gift cards don’t need it—they work automatically.” [Interviewee #9]

Some progress has been made via interface of SCOs in payment steps for S Group stores, such as visual animations that guide customers to press “Pay” or confirm card status.

“Nowadays our monitor says it’s time to pay and there’s an animation... that has helped a lot.” [Interviewee #5]

To reduce the need for staff intervention, the payment process should be simplified in some implementable ways. First, removing redundant steps, such as requiring customers to manually press “Proceed to Pay” after scanning a loyalty card, would streamline checkout and reduce confusion. Second, allowing customers to retry failed payments without staff assistance would resolve most issues independently. These changes are technically simple and widely applicable. In contrast, achieving consistent gift card handling is more complex, as it involves coordination with third-party systems and financial providers. Still, together, these improvements would make SCOs more autonomous and reduce unnecessary interruptions for both customers and employees.

4.1.7 Order Cancellation

Order cancellation at SCOs is a recurring disruption that requires staff assistance. Customers can scan items but then realize they cannot complete the payment due to various reasons such as missing a payment card, wanting to pay with cash, or simply changing their mind. With SCOs, they cannot cancel the transaction themselves. Instead, they must call for help from

employees, or worse, leave the checkout incomplete, which blocks the machine for the next user. This manual intervention slows down the line, creates confusion, and adds to employee workload. Customers cannot currently cancel the transaction themselves. One reason for this limitation is concerns about theft. If customers could freely cancel their order, they might cancel and leave with unpaid products undetected.

“Often, when customers realize they can’t pay with cash or don’t have their card, they leave the self-checkout without saying anything... this causes issues for other customers... staff then need to step in and clear the abandoned transaction.” [Interviewee #4]

“If a customer scans items at self-checkout but then decides to pay with cash, they can’t cancel the order themselves and need employee assistance.” [Interviewee #5]

Although some stores allow remote cancellation from employee monitors, this still requires employee attention and does not empower the customer to resolve the issue independently.

“I can also, like, cancel the whole purchase that was earlier made by that person, like, on my own screen, so then the line can continue to the self-service checkout.” [Interviewee #3]

“If they change their mind and choose to come to the normal cashier, I can cancel the order from the self-service checkout now remotely.” [Interviewee #6]

While this remote control is helpful for employee efficiency, it does not solve the core issue. Customers cannot cancel their own transactions, especially when they abandon the SCO without informing anyone and leave the system blocked. In stores with receipt-activated exit gates, this concern can be addressed. These gates require a printed receipt to open, effectively ensuring

that no customer can leave the area without completing payment. In such environments, allowing customers to cancel their order from the screen would be a safe and efficient improvement.

“If you just cancel your items and do not have the receipt, then you cannot go from the gates. So it would be nice to have the cancel option also [at the machine].” [Interviewee #6]

Providing customers with the ability to cancel their own transaction safely could prevent bottlenecks, reduce unnecessary staff workload, and ensure smoother traffic flow in gated self-checkout area. However, in stores without exit gates, cancellation must still be handled manually by staff to ensure customers do not leave the area with unpaid goods.

4.1.8 Missing Price Tag

A recurring challenge at SCOs is the lack of price information, particularly with newly stocked or unlabelled items. These items often trigger a “call for assistance” notification, as the system cannot recognize the product or its EAN code. From the scope of SCO usage, there is nothing much to improve when the bar code is not in the system for newly stocked or unlabelled items.

“Another frequent issue is the ‘call for assistance’ notification when a product is missing a price. This often happens when new products are stocked but their prices have been forgotten. EAN codes sometimes change even if the product looks the same, which means new prices need to be added manually.” [Interviewee #2]

This issue also overlaps with challenges around weighing fresh produce, where customers frequently forget to weigh their items before reaching checkout. Interviewee 4 described the impact on operations:

“Employees must either instruct customers to reweigh items or manually weigh them using a small scale at the cashier area, which can

be slow—especially if the item code is missing or unfamiliar.”
[Interviewee #4]

In self-checkout, this often causes timeouts when customers leave to weigh forgotten items. During rush hours, such issues make delays even worse.

“Customers often forget to weigh their produce before reaching the cashier or self-checkout... they might leave their items to weigh something and trigger a timeout error. Staff sometimes fix this quietly, but it slows down operations.” [Interviewee #6]

This problem can be addressed by integrating scales directly into SCOs, where customers can weigh items, select them from an on-screen menu, and print a barcode label. Interviewee 4 pointed out that this model works well in smaller stores. However, in large stores with more product variety, such as Prisma and K-Citymarket, customers may struggle to identify or name items correctly.

“Some stores like Lidl have integrated scales into self-checkouts, allowing customers to weigh produce and print labels there, which speeds up the process. However, this may not work as well in large stores like Prisma, which have many similar produce items, making it harder for customers to remember what they picked.” [Interviewee #4]

“The best solution may be to have both options available: integrated scales at checkout machine and regular scales in the self-service area.”
[Interviewee #6]

It is also noted that placing a scale near the checkout could streamline the process and reduce delays for both staff and shoppers. Customers or employees can save time by avoiding an extra trip to the produce section that often has the items needing to be weighed.

“[There should be] a scale in the self-checkout area - the scale that can generate the price tag. The scale can have interface to help users identify their products” [Interviewee #7]

Together, these insights suggest that improving price tag accuracy, integrating weighing systems into SCO machines, and offering scale options would significantly reduce disruptions for both customers and employees at checkout.

4.2 Process and Tech Improvements

This section outlines practical improvements that address both common SCO problems and overall system usability. Some solutions directly reduce staff intervention and prevent user frustration, while others serve as enhancements to improve flow, confidence, and accessibility for all customers. These improvements are grouped into three themes: interface design, remote assistance, and clearer communication.

4.2.1 Better Interface Design

A user-friendly interface is crucial in minimizing customer mistakes in SCO systems. Multiple interviewees in this study emphasized that many common issues, such as accidental multiple scanning, bonus card confusion, and confusing sequence of actions, come from unintuitive interface design rather than customer mistakes.

This aligns with research by Cho & Fiorito (2010), who argue that poorly designed SCOs often lead to process interruptions. Similarly, Galdolage (2022) found that when customers have limited technological proficiency, even small interface flaws can quickly escalate into service.

Interviewees reported that certain interface elements were overly sensitive or lacked feedback clarity. For example, the inability to edit item quantities

directly leads to customer frustration and frequent staff assistance. Several employees suggested adding quantity adjustment buttons along with a distinct audio sound. This improvement allows staff to distinguish between scanning and editing actions from a distance. While this introduces a minor security concern, interviewees generally agreed that a proper sound would maintain supervision.

Timeout notifications represent another common source of confusion, particularly among elderly or distracted users. Current systems misinterpret slow behavior as abandonment, often triggering unnecessary staff interventions. Interviewees recommended adding a “More Time” button that would allow users to self-extend their session, along with a subtle sound or visual confirmation for staff. This low-cost interface fix would reduce disruptions, maintain flow, and prevent customers from feeling rushed or penalized.

As Zahari et al. (2023) highlight in their SST study, customers often fail to follow interface steps due to poor screen design or unclear instructions. For instance, in our interviews, the “Proceed to Pay” step that is required even after scanning the loyalty card was frequently missed, especially by elderly customers. One interviewee contrasted this with S Group kiosks, which automatically continue to payment after loyalty card recognition, reducing both confusion and timeout errors. This exemplifies what Bitner et al., 2002 advocate as “seamless flow design”, where fewer steps lead to lower friction and higher user satisfaction.

Another example includes the lack of dynamic language switching mid-checkout. Although kiosks offer language selection at the beginning, users who mistakenly press the wrong language cannot correct it without starting over, which can be a significant barrier for non-Finnish speakers. This aligns with (Nili et al., 2019), who emphasize the value of adaptive interfaces that respond to user errors in real time.

In summary, better interface design does not only improve efficiency; it also enhances user autonomy and reduces the invisible burden on staff. As Dabholkar & Spaid (2012) argue, self-service experiences succeed when customers feel in control and confident navigating the technology, especially in the absence of immediate assistance.

4.2.2 Enhanced Remote Assistance

Remote assistance capabilities have emerged as a valuable innovation in streamlining SCO operations. When implemented effectively, these systems allow employees to monitor and intervene in SCO transactions from a cashier station. This functionality enables staff to resolve common issues such as accidental double scans, timeout alerts, and approval of age-restricted products without physically presence the machines. As a result, both employee workload and customer waiting times are significantly reduced.

This approach reflects the broader trend toward automation-supported service recovery. This concept is discussed by Bitner et al. (2002) who emphasize the role of technology in enabling efficient “backstage” problem resolution. Similarly, Dao & Theotokis (2021) argue that recovery processes initiated by the system or employees without disrupting the user flow can preserve service satisfaction even during technical or process failures. Remote monitoring aligns with this by allowing employees to stay available while minimizing interruption.

Smaller convenience stores have been the first to adopt these tools due to their compact layouts and multitasking demands. In large hypermarkets, dedicated staff are usually stationed at SCO areas, making remote access seem less critical. However, Mattila & Cho (2011) suggest that even in high-traffic environments, empowering support staff with remote capabilities increases their capacity to manage multiple kiosks, especially during peaks or shift gaps.

Moreover, remote assistance systems promote employee wellbeing by reducing the physical strain of frequent walking between machines, which is a concern in many studies on technology-mediated service work (Nili et al., 2019). Remote access tools should expand to broader intervention, such as applying discounts or modifying prices. Remote assistance also supports the principle of “invisible recovery,” where customers are unaware of fixes, but still benefit from a seamless experience (Meuter et al., 2000).

Ultimately, enhanced remote assistance improves operational efficiency and employee wellbeing. As SCOs continue evolving, remote support systems can be integrated for a more autonomous retail environment.

4.2.3 Clearer Communication

Beyond interface design, pre-checkout guidance plays a critical role in shaping customer behavior and minimizing service failures at SCO machines. Multiple employees emphasized that many avoidable interruptions could be significantly reduced with clearer physical signage and more intuitive instructions before users begin the checkout process.

Research by Bitner et al. (2002) highlights that self-service environments require user education to reduce abandonment and employee dependency. When customers enter the SCO area with incomplete understanding, small friction points become potential failure. In practice, signs placed too close to the kiosk, or using dense text, are often missed or misunderstood. This is particularly problematic in high-traffic environments or for customers unfamiliar with local retail norms (B. S. Galdolage, 2022).

Interviewees recommended large, simple, and visual signage at the entrance to the SCO area that clearly communicates key rules: for instance, “Pantti receipts must be processed at the cashier,” or “Only barcode-labeled discounts can be used here.” As one interviewee noted, even signs that do

exist are often too text-heavy or poorly located to be effective. Aligning with findings by Meuter et al. (2000), this reinforces the need for icons, visual stickers, and even color-coded lighting to act as passive guidance systems, especially for non-native speakers or elderly customers.

Audio cues were also highlighted as important complements to static signs. For example, distinct sounds could confirm when items are removed or edited, providing feedback to both users and staff. This concept, supported by Forbes, (2008), reflects a growing emphasis on ambient awareness tools such as non-intrusive cues that help staff monitor kiosks without interrupting users. Likewise, short but eye-catching reminders for common errors (“Please press 'Pay' to continue”) can guide users gently without requiring staff intervention.

In short, improving SCO effectiveness starts before a customer even touches the screen. High-contrast and image-based signage, color-coded visual guides, and distinct audio cues all contribute to more autonomous user behavior. Together, these can reduce confusion, speed up checkout time, and lighten the burden on employees.

4.3 Balancing Efficiency and Empathy in SCO Support

In SCO environments, employees face the challenge of maintaining operational efficiency while providing empathetic, customer-oriented support. This balance becomes even more complex because SCOs are designed to reduce human intervention, but also require frequent staff involvement due to user errors, system limitations, and customer anxiety, especially among the elderly or those unfamiliar with the interface.

4.3.1 Emotional Labor and Invisible Workload

Employees are expected to be alert, calm, and helpful with customers who are frustrated or confused. This aligns with research on emotional labor in retail, which shows that service workers must control their emotions to create a pleasant customer experience (Brotheridge & Grandey, 2002). Several

interviewees highlighted the stress of managing long lines of customers waiting for SCOs and helping with errors.

"It's really embarrassing when there's a long line and you're stuck helping one customer with something small. They [customers] think we're ignoring them." [Interviewee #4]

"There is situation where there's a long line and the customer doesn't know what to do, so we try not to make them feel bad, just fix the problem fast" [Interviewee #9]

Employees not only solve SCO problems but also manage customer moods and confusion, while trying to appear efficient and friendly. These tasks might seem minor but carry emotional weight. Literature on service work supports this: Hochschild (2002) introduced the concept of emotional labor to describe how workers manage their feelings and expressions to meet job expectations. In SCO settings, where technological breakdowns or user confusion are common, this labor is not often recognized but is essential to maintain customer satisfaction and loyalty (Giebelhausen et al., 2014).

4.3.2 Random Checks and Trust

A unique operational feature in many K Group stores, not present in S Group stores, is the mandatory random check at SCOs. These checks occur even when the machine finds no errors, and they require employees to inspect a customer's scanned items before allowing them to proceed. While designed to detect theft and ensure scanning accuracy, the process introduces a confusing interaction that is often interpreted by customers as suspicion.

Several employees mentioned that customers appear visibly annoyed when selected for a random check because they feel they have done everything correctly.

"People seem a little annoyed at first, but I just explain it's random, nothing personal. I keep it light and fast, and usually they relax and

sometimes they even smile. I think it's nice to have a moment to interact with the customer.” [Interviewee #6]

This perspective illustrates how employees can transform a potentially negative moment into a positive one. The same interviewee also said that they value the opportunity to talk to customers and maintain a sense of human presence in an automated process. Still, another employee expressed discomfort with the practice, especially when they felt it interrupted the flow of service or caused unnecessary tension.

“You have to randomly check their items even when everything goes fine. I understand why sometimes people take it really personally.” [Interviewee #3]

These comments highlight the emotional labor involved in enforcing store policy while also trying to maintain customers' mood. Employees must reassure customers that the check is not a sign of mistrust while moving quickly to minimize disruption. This aligns with research by (Giebelhausen et al., 2014), which found that employees who manage emotional situations effectively can lessen the negative effects of interruptions on customer satisfaction. From an operational standpoint, these interactions may slow down overall checkout process. However, the employee-customer relationship can be fostered with trust and loyalty.

4.3.3 “Roaming Assistance” and the Physical Burden

Interviewees in small stores without an assigned employee at the checkout area emphasized how physically demanding it is to constantly move to the area to provide support, especially when they are doing something elsewhere.

“I assist the SCO every five minutes and just for small things customers can't fix. It interrupts other tasks like stocking” [Interviewee #2]

Some small stores use remote assistance systems to reduce this burden.

“We can now solve some things from the cashier station, like clearing timeouts or approving ID. This saves a lot of walking” [Interviewee #3]

Still, these tools are not universally available since they are still in testing phase. Based on employees’ perspective, remote assistance can only be used in small stores such as K-Market, S-Market or Alepa. In some larger stores, an employee is assigned at the SCO area solely to help at self-checkouts, a role that increases visibility but also amplifies the pressure to resolve issues rapidly and kindly.

4.3.4 Cultural Context and Communication Barriers

Employees consistently reported that language and communication gaps often create challenges at SCOs, particularly for elderly customers and international customers unfamiliar with the Finnish retail system. Interviewees mentioned common misunderstandings such as customers failing to press “no customer card” before attempting to pay, or being unsure about how to weigh produce or handle pantti receipts. While most SCOs provide language options, customers must choose their language at the beginning of the process. Several employees pointed out that customers who accidentally press the wrong language cannot change it mid-checkout, which can lead to confusion and frustration.

“Many people just try to scan their card without pressing anything. Especially older people or those not fluent in Finnish get confused easily even with English option.” [Interviewee #9]

This problem reflects a broader challenge of designing user interfaces that accommodate diverse linguistic and digital literacy backgrounds. Some stores use animations and visual cues, which have helped reduce these issues.

“Our checkout now has an animation saying, ‘It’s time to pay,’ and that’s helped a lot. Every store has this kind of instruction but maybe

it doesn't stand out enough to catch customers' attention."
[Interviewee #5]

From these observations, service design needs to be inclusive design to support all user groups (Lindsay & Norman, 2013). Clear visuals, multilingual prompts, and intuitive flow are very important. Some interviewees mentioned that customers may feel embarrassed when making mistakes, which leads them to adopt "silent frustration." This means customers hesitate to ask for help. This emotional aspect is common in Finland's low-contact service culture, where autonomy is highly valued. Employees adapt by remaining observant and only intervening when necessary.

"Sometimes customers just stand there looking confused, but they won't ask for help. You have to notice it yourself and go offer support without making them feel embarrassed." [Interviewee #4]

Interviewees agreed that clearer, more flexible language settings and visual cues are essential for reducing reliance on employee intervention. This not only supports customer autonomy but also lessens the emotional and time burden on employees, particularly during peak hours.

4.3.5 Empathy Slows Down Efficiency but Increases Satisfaction

Interviewees also mentioned that supporting elderly users or anxious customers often slows down the process but enhances customer experience. Staff feel pressure to act fast, but when they take a moment to calm a frustrated customer, the overall outcome is more positive. This aligns with Dabholkar & Spaid (2012) findings, which emphasize the importance of interpersonal trust and emotional safety in technology-based services.

"You have to be patient as some people need more time or want to say something before paying." [Interviewee #9]

“Some customers are slow or stressed. I just explain calmly and smile, and it makes them feel better,” [Interviewee #7]

“Even when people seem angry, it’s often because they don’t understand the system. You just explain, and they thank you” [Interviewee #4]

While retail KPIs may prioritize the efficiency, employees recognize that human connection can be a competitive advantage. Especially in grocery settings, customers often return to stores where they feel understood. This reflects the trade-off between speed and service quality. Research by Parasuraman et al. (2005) highlighted that emotional support is a key dimension of service quality in tech-based service environments. While automation increases speed, human empathy still defines the perceived quality of the experience.

Balancing efficiency with empathy in SCO support is a demanding but critical task. Employees need to solve SCO errors efficiently while constantly managing customer expectations. While automation may reduce routine tasks, it cannot eliminate the human dimension of retail work. As future SCO designs evolve, system developers and managers should prioritize tools that reduce unnecessary friction while empowering employees to focus on meaningful and supportive interactions.

4.4 Main Findings

This section summarizes the key findings based on the two research questions: (1) What common problems do customers face with SCOs that need employee help? and (2) How can the design, technology, or instructions of SCOs be improved to help customers resolve these failures independently, reducing the need for employee intervention?

The following table synthesizes the key findings from Section 4.1 and 4.2. The analysis of employee interviews and operational observations revealed

several key pain points that commonly disrupt the SCO experience. Each of these challenges contributes to increased customer confusion, longer transaction times, and greater staff workload due to necessary interventions. The following table synthesizes these main findings by describing the problem, its root causes, and suggested improvements that could enhance the autonomy and efficiency of SCOs.

Issue	Description	Root Causes	Solutions
Item Quantities and Scanning Mistakes	Accidental double-scanning of items; customers cannot remove items themselves	Lack of user control; theft risk concerns	- Allow customers to delete items - Use a distinct alert sound when items are removed - Implement remote monitoring so staff can verify changes - Introduce audit trails or confirmation prompts
Alcoholic and Age-Restricted Products	Frequent staff intervention required to verify age	Legal requirement; system lacks digital verification	- Enable remote approval interfaces with judgment buttons - Integrate digital ID verification at SCOs
Timeout Notifications	Sessions timeout too quickly; system misinterprets slow actions as abandonment	Short default timeout; no way for customer to respond	- Add “I need more time” button - Extend default timeout - Retain remote clearing capability by staff
Bottle Deposit Refund (Pantti)	Pantti receipts cannot be processed by customers at SCOs	SCO systems cannot validate single-use receipts	- Enable barcode validation for pantti receipts - Add clear signs to explain usage - Enhance UI to prompt user correctly
Discounted Items or Coupons	Inconsistent recognition of discounted products	Non-standardized labels; wrong	- Use barcode-based discount stickers with new EANs - Ensure proper placement of stickers

		barcode scanned	- Train customers via signage or prompts
Payment Interface Friction	Customers get confused after scanning loyalty cards because they must manually press "Proceed to Pay". Payment retries also require staff assistance.	Redundant confirmation step; lack of autonomy in retrying failed payments; limited gift card compatibility.	- Auto-advance to payment after loyalty card scan. - Allow customer-initiated payment retries. - Improve visual guidance. - Standardize gift card handling to reduce staff interventions.
Order Cancellation	Customers cannot cancel their own order; they must call for staff, or leave without informing, blocking the SCO and delaying others.	No customer-facing cancellation option due to theft concerns.	- Enable customer cancellation in gated SCO setups. - Allow remote employee cancellation. - Add "Cancel Order" option on-screen in secure environments.
Missing Price Tag	Items without price labels (especially fresh produce or new stock) trigger assistance requests and checkout timeouts.	Barcode not in system; produce not weighed or labeled in advance.	- Integrate scales at SCOs. - Allow barcode label printing at checkout. - Place additional scales near SCO area. - Improve product identification interface for produce selection.

Table 4: Summary of common SCO errors and their solutions

Together, these findings suggest that the challenges faced at SCOs are not due to technological failure, but to gaps in user-centered design, interface clarity, and communication. By addressing these gaps, stores can

significantly reduce the frequency of staff interventions, improve customer autonomy, and enhance the overall self-checkout experience.

The following table synthesizes the key findings from Section 4.3. It highlights the emotional and physical demands placed on employees, the complexities of enforcing policies like random checks, and the communication barriers that affect diverse customer groups.

Theme	Description
Emotional Labor & Invisible Workload	Employees must manage their own emotions and customer moods under pressure of store tasks and SCO errors
Random Checks & Trust	Mandatory random item checks (in K Group only) often trigger customer annoyance or suspicion, but skilled employees can diffuse tension and even build rapport during these brief interactions.
Roaming Assistance & Physical Burden	Constantly moving to assist scattered SCOs interrupts other tasks. Remote-assist tools help in small stores but are not available in larger stores.
Cultural Context & Communication Barriers	Language selection errors and interface misunderstandings (especially among elderly or non-Finnish speakers) lead to silent frustration. Visual cues and flexible language toggles help but need to be more prominent and adaptive.
Empathy Slows Efficiency but Increases Satisfaction	Taking extra time to calm or guide customers (particularly elderly or anxious) may slow checkout speeds but leads to higher perceived service quality and repeat visits.

Table 5: Findings for balancing efficiency and empathy in SCO support

5 Discussion

5.1 Theoretical contribution

This study contributes to the growing body of literature on technology-based service encounters by offering an employee-centered perspective on SST challenges and design implications. While previous research has often prioritized customer satisfaction, adoption intentions (Bitner et al., 2002; Meuter et al., 2000; Tapani Rinta-Kahila, 2013), and service recovery strategies (Collier et al., 2017; Dabholkar & Spaid, 2012), this thesis focuses on the operational realities faced by retail employees in managing SCO-related errors. This study also reveals the invisible emotional labor behind SCOs, which are often seen as autonomous but still rely heavily on employee support. It highlights how employees must manage customer confusion, frustration, and system errors while maintaining speed and empathy; this is often overlooked in SST research. It also extends emotional labor theory into tech-driven retail, offering a more human-centered view of service quality.

By identifying and categorizing recurring SCO errors and their triggers, this study extends the typology of SST failure scenarios (B. S. Galdolage, 2022; Nili et al., 2019) and emphasizes the interplay between customer interface design, system limitations, and real-world employee interventions. The findings illustrate how certain technical designs create bottlenecks not just for customers, but for store operations. These insights support the argument made by (Zahari et al., 2023) that SST design must reflect user diversity, not just in technological proficiency, but also in physical and cognitive ability.

Importantly, this thesis contributes to the Finnish retail technology knowledge, which has seen rapid SCO adoption across chains like K Group and S Group. It gives context on how unique local factors - such as pantti systems, gate-restricted SCO zones, and a linguistically diverse customer base - shape employee intervention patterns and user experience. By doing so, it fills a gap in SST research, which often overlooks country-specific implementations and the cultural aspects of the interaction.

Ultimately, this thesis offers a grounded, system-level perspective on how SCOs could evolve to better balance automation and human support. It reinforces theoretical frameworks of service recovery, perceived control (Chiu & Nguyen, 2022), and customer autonomy. It is suggested that successful SCO deployment requires not only technological functionality but also adaptability, clarity, and empathy in design.

5.2 Managerial implications

The findings of this thesis highlight several important managerial considerations for retailers operating in the Finnish grocery sector, where SCOs are now a common solution.

- SCOs remain labor-intensive despite the intent to reduce staffing needs; employee assistance is still frequently required due to system limitations and unclear customer processes.
- Usability improvements to SCO interfaces are one of the most direct ways to reduce disruptions. Even small design changes can significantly benefit user experience, especially for elderly or unfamiliar users.
- Clear, multilingual signage at key points in the checkout journey is critical. Many customer errors come from lack of visible instructions or poor process guidance.
- Remote assistance technologies are highly effective in smaller stores. Tools that allow staff to assist SCO users without physical presence increase efficiency and reduce employee fatigue.

- Employee feedback should be systematically integrated into technology updates and SCO management. Employees can help identify recurring problems and offer practical, low-cost solutions.
- SCOs should be managed as hybrid service systems, not just cost-cutting tools. A successful SCO strategy combines automation with thoughtful service design, employee support, and iterative improvement.
- Customer autonomy and satisfaction should be prioritized alongside operational efficiency. When both customer and employee needs are addressed, SCOs are more likely to deliver their intended value.

5.3 Limitations and future research

While offering practical and theoretical insights into SCO use in Finnish grocery retail, this study also has several limitations that should be acknowledged. First, the data were collected entirely from the perspective of store employees. Although this provided valuable insights into the operational challenges and recurring intervention points at SCOs, it did not capture the customers' direct experiences, expectations, or frustrations. Future research could complement these findings by including customer interviews or surveys to explore the usability of SCO systems from the customer perspective, particularly across different demographic groups such as elderly shoppers, immigrants, or first-time users.

Second, the scope of this study was limited to a select number of stores in Finland, within K Group and S Group stores. Because these stores vary in size and operational context, the findings may not fully represent other chains or international retailers operating in Finland.

Another limitation concerns the rapid pace of technological change. Several features discussed in this study, such as remote cashier assistance or digital ID verification, are currently being piloted or partially rolled out in select locations. As these systems evolve and become more widely adopted, the types of errors and employee roles associated with SCOs may shift significantly. Long-term research would be valuable to track how new technologies affect customer autonomy, employee workload, and operational efficiency over time.

Lastly, this study focused primarily on problem-solving and recovery within the checkout process. It did not explore other factors that may influence SCO success, such as employee training practices, store layout, or customer onboarding for self-service tools. Future research could examine how organizational support structures contribute to the effectiveness of self-service systems.

In summary, since this thesis only focuses on key barriers and potential improvements to SCO usage from the employee perspective, there is opportunity for future research to broaden the scope, include multiple viewpoints, and track the evolving relationship between people and self-service technologies in retail environments.

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Appendices

Appendix 1: Interview Questions

Self-Service Checkout (SSC) Failures & Customer Self-Resolution

Thank you for taking the time to participate in this study. Your insights will contribute to research aimed at improving self-service checkouts (SSCs) for both customers and employees.

This study focuses on identifying common SSC failures that customers could resolve independently, reducing the need for cashier intervention. From the employee perspective, the objective is to understand:

- Which SSC failures frequently require assistance.
- Why employees believe customers could resolve these failures themselves.
- How enabling customer self-resolution impacts employee efficiency and workload.

By gathering your firsthand experience, this research seeks to propose solutions that enhance the SSC experience, reduce unnecessary disruptions, and improve overall store efficiency.

Consent & Anonymity: Your responses will be kept confidential and used solely for academic research in my thesis. No personal or identifiable information will be included in the final report. However, I will refer to your store size and company when analyzing the data. If you agree to participate, please answer the questions based on your experience.

Language Simplicity: Please answer based on your daily experience. There are no right or wrong answers—your insights from working with SSCs are highly valuable.

1. Background & Experience

- How long have you worked at this store?
- What is your role, and how often do you assist customers at SSCs?
- How large is your store? (e.g., small convenience store, mid-sized supermarket, large hypermarket)
- How many SSCs does your store have? Are they widely used by customers?
- What are the common SSC issues that you need to assist the customers? This includes all types of SSC issues requiring employee assistance, regardless of the cause (e.g., technical malfunctions, customer mistakes, system limitations).

2. SSC Failures & Customer Self-Resolution

Errors requiring unnecessary cashier intervention

- Which SSC errors currently require your help but could be designed for easier self-resolution?
- What are the most frequent mistakes customers make due to unclear SSC instructions or error messages?

- Are there common pricing issues (e.g., incorrect discounts, missing promotions) that customers could resolve without manual intervention?
- Do customers often get stuck due to scanning, bagging, or payment issues? Which of these could be made easier for them to handle independently? (take too long to scan the bonus card,...)

3. Technology & Process Improvements

- What SSC features or changes would reduce the need for cashier intervention?
- If you could improve one thing about SSCs to make them more user-friendly, what would it be?

4. Final Thoughts

- Is there anything else you'd like to share about SSC challenges or potential improvements in general? Maybe something that I have not asked above.